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FEATURE ARTICLE

Molecular phylogenetic and population genetic relationships of a putative species of sucker (*Catostomus* sp.) from Surprise Valley in the Great Basin, USA

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Abstract

Objective: The ichthyofaunal diversity of the Great Basin is incompletely characterized, with the Wall Canyon Sucker *Catostomus* sp. being one potential species. The Wall Canyon Sucker is limited in distribution to the Wall Canyon drainage of Surprise Valley, Washoe County, Nevada, USA. It was first documented and collected by C. L. Hubbs and R. R. Miller in 1934, and no species description has been published to date. Nonetheless, the Wall Canyon Sucker has been treated as a species and has appeared in various contexts within and outside of the scientific literature as such.

Methods: We conducted phylogenetic analyses and the first population genetics study of this putative species using genome-wide sampling obtained through restriction site-associated DNA sequencing to assess the Wall Canyon Sucker's relationships and population genetic characteristics.

Result: We found the Wall Canyon Sucker to be sister to the Warner Sucker *C. warnerensis* with high support and a median divergence time of 2.12 million years. Population genetic analyses supported that the Wall Canyon Sucker is a single genetic population with a high degree of isolation. Principal components and admixture analyses did not indicate recent admixture of the Wall Canyon Sucker with the Warner Sucker. Substantial differences in allele frequencies between the Wall Canyon and Warner suckers were apparent (genetic differentiation index $F_{ST}=0.15–0.21$). Nucleotide diversity in the Wall Canyon Sucker was the lowest of all observations, and Tajima's D was greater than zero ($D=1.96$), indicating population contraction and a lack of rare alleles, congruent with the desiccation of Lake Surprise and the persistence of the fish in a very restricted recent range. We undertook species delimitation with publicly available mitochondrial gene sequence data and found that a species-level designation of the Wall Canyon Sucker is appropriate in that framework as well.

Conclusion: The results of this study are consistent with recognition of the Wall Canyon Sucker as a species; however, it still requires a formal taxonomic description.

KEYWORDS

candidate species, conservation genetics, molecular species delimitation, RADseq

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INTRODUCTION

The Wall Canyon Sucker *Catostomus* sp., also known as the Surprise Valley Sucker, is restricted to Wall Canyon Creek and associated drainages in northwestern Nevada, USA (Figure 1A). The Wall Canyon Sucker was first documented by C. L. Hubbs and R. R. Miller in 1934, and these experts recognized the sucker as a new species (Miller et al. 1991). To date, a formal species description remains unpublished. Despite the lack of a species epithet, the Wall Canyon Sucker still appears in various contexts as a species, such as in scientific manuscripts (e.g., Siebert and Minckley 1986; Bagley et al. 2018) and Endangered Species Act documentation (U.S. Fish and Wildlife Service 1994).

Siebert and Minckley (1986) indicated that the Wall Canyon Sucker belongs to a broadly distributed clade from Oregon to northwest Mexico based on "resemblances in scalation, body shape, pigmentation, and lip structure." They considered the Modoc Sucker *Catostomus microps* and other suckers of the genus *Catostomus* of the northern Sacramento River and nearby drainages—the Tahoe Sucker *C. tahoensis*, Warner Sucker *C. warnerensis*, and Owens Sucker *C. fumeiventris* (Figure 1A)—to be lineages closely related to the Wall Canyon Sucker. Molecular phylogenetic hypotheses regarding suckers have demonstrated the non-monophyly of *Catostomus* and that the genera *Chasmistes*, *Deltistes*, and *Xyrauchen* may be synonymized with *Catostomus* (e.g., Chen and Mayden 2012). This result strongly suggests that correlating morphological divergence with time of divergence within the

Impact Statement

Lack of taxonomic recognition is a major shortfall in conservation. We investigated a putative species, the Wall Canyon Sucker *Catostomus* sp., with molecular phylogenetic and population genetic methods. Our results were consistent with the Wall Canyon Sucker being a species-level lineage.

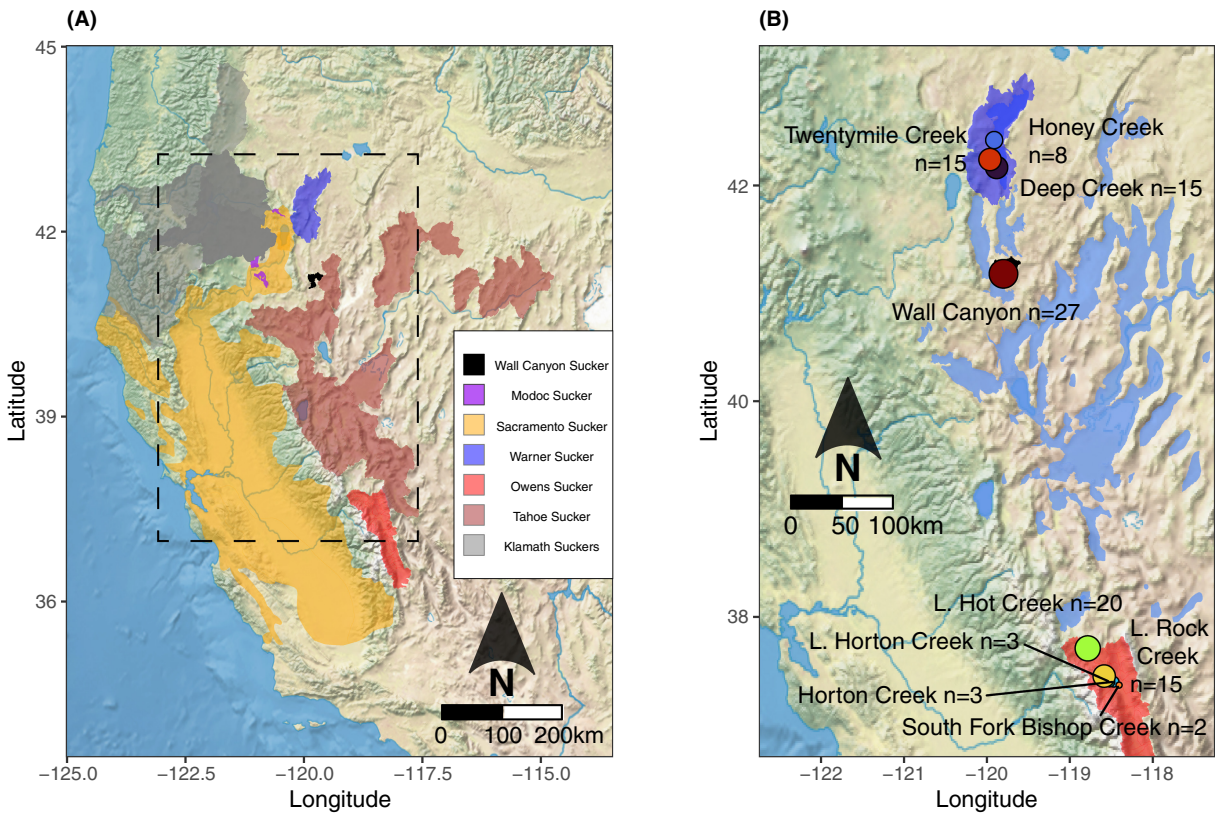

FIGURE 1 (A) Distribution of key sucker (Catostomidae) lineages examined in this study. Klamath suckers include multiple sublineages that are not shown individually (Klamath Largescale Sucker *Catostomus snyderi*, Klamath Smallscale Sucker *Catostomus rimiculus*, Shortnose Sucker *Chasmistes brevirostris*, and Lost River Sucker *Deltistes luxatus*). The distribution of the Lahontan Sucker *Pantosteus lahontan* is not shown. The dashed outline shows the extent of panel B. (B) Sampling locations for the Wall Canyon Sucker *Catostomus* sp., Warner Sucker *Catostomus warnerensis*, and Owens Sucker *Catostomus fumeiventris* are indicated with pluvial lakes of the western Great Basin (https://earthworks.stanford.edu/catalog/stanford-xb000vk8400). Each point represents a sampling location and is uniquely colored. The size of points is proportional to the number of individuals sampled. "L." indicates Lower.

Catostomidae and invoking complex hybridization scenarios to explain observed molecular phylogenetic relationships require re-evaluation in light of new evidence (Chen and Mayden 2012). Molecular phylogenetics have also indicated that mountain suckers *Pantosteus* spp., often considered a subgenus of *Catostomus*, merit recognition as a separate genus with higher than previously recognized species diversity (Unmack et al. 2014; Bagley et al. 2018). *Pantosteus* is the sister lineage of *Catostomus* (including *Chasmistes*, *Deltistes*, and *Xyrauchen*) and is an appropriate outgroup for molecular phylogenetic studies of *Catostomus*.

The most comprehensive molecular phylogenetic hypotheses of sucker relationships proposed by Bagley et al. (2018) largely support the closely allied species of suckers to the Wall Canyon Sucker as indicated by Siebert and Minckley (1986). The Wall Canyon and Warner suckers are sister taxa that, together, are most closely related to the Owens Sucker. An estimated divergence time for the Wall Canyon Sucker and the Warner Sucker has a lower 95% highest posterior density (HPD) that is pre-Pleistocene (~2.58 million years ago [mya]). The Wall Canyon, Owens, and Warner suckers are near-relatives to lineages from the Klamath Basin (including species of the genera *Chasmistes* and *Deltistes*), the Modoc Sucker, and the Sacramento Sucker *Catostomus occidentalis*.

The relationships shown by Bagley et al. (2018) are largely driven by variation in mitochondrial DNA (mtDNA) and are analyzed in a concatenated framework. The findings are informative but are only based on a few individuals (e.g., $n = 1$ in fossil-calibrated analyses of both the Wall Canyon and Warner suckers) and do not explicitly test whether the Wall Canyon Sucker is an independent species. Rather, the divergence time reported by Bagley et al. (2018) was calculated with the assumption that the Wall Canyon Sucker is an independent lineage, and the concatenated phylogenetic frameworks did not accurately represent hybridization or gene flow. Furthermore, Bagley et al. (2018) applied a total-evidence divergence time estimation approach for suckers. Under this framework, fossils are placed in the phylogeny with a matrix of characters, and the associated dating information is used to time-calibrate the tree. These combined data generate topologies that are substantially different from molecular-only topologies as a result of the conflicting signal surrounding the monophyly of *Catostomus*. Such an approach may result in an altered estimate of divergence time of the Wall Canyon Sucker from the Warner Sucker when compared to estimates obtained through other methods. Current publicly available data lack substantial individual or genomic sampling for population genetic metrics, and the Wall Canyon Sucker's degree of isolation is difficult to gauge. The effect of hybridization on phylogenies is notable because it is a phenomenon that has been documented in suckers of the Klamath Basin (Dowling et al. 2016) and between the Modoc and Sacramento suckers (Smith et al. 2011). Population genetic characterization can reflect this and other processes that are not apparent in concatenated phylogenetic analyses.

The claim of species-level standing for the Wall Canyon Sucker remains unsubstantiated, and it may represent post-pluvial isolation that would be better characterized as an isolated population of an existing described taxon. The recognition of the Wall Canyon Sucker as a subspecies of a more broadly distributed sucker lineage may also be a possible resolution to the current taxonomic ambiguity of the Wall Canyon Sucker. Alternatively, the Wall Canyon Sucker may be, in fact, an undescribed species-level taxon. We re-assessed the phylogenetic relationships of the Wall Canyon Sucker and divergence times of *Catostomus* without assuming the monophyly of *Catostomus*. We also examined the phylogenetic and population genetic relationships and characteristics of the Wall Canyon Sucker and potential near-relatives with increased sampling of individuals and genetic loci (restriction site-associated DNA sequencing [RADseq]). With this increased sampling, we investigated the species-level standing of this lineage, and we provide a genetic baseline for future studies focused on evaluating and monitoring this candidate species. Finally, we applied a molecular species delimitation method to test whether the Wall Canyon Sucker is a species in that framework.

MATERIALS AND METHODS

Re-assessment of *Catostomus* relationships and divergence times

We obtained the original alignment from Bagley et al. (2018), consisting of 85 total organisms with 80 catostomids, which was based on mitochondrial cytochrome b (*cytb*), mitochondrial NADH subunit 2 (*nd2*), mitochondrial cytochrome c oxidase 1 (*cox1*), interphotoreceptor retinoid-binding protein (*irbp*), ribosomal protein S7 intron 1 (*rps7* I), and growth hormone intron (*gh* I) loci. The mitochondrial loci may be considered a single locus, thus resulting in a total of four loci.

We searched GenBank with the Basic Local Alignment Search Tool (BLAST) for near-relatives of the Wall Canyon Sucker that may not have been previously identified due to the assumption of *Catostomus* monophyly. We added sequences to the existing alignment of Cui-ui *Chasmistes*

cujus from *cytb* (KU697934.1) and *nd2* (KU697962.1) mitochondrial genes as a result.

We reduced the number of species in the alignment by removing non-catostomids and suckers from outside of the Catostomini (i.e., those not belonging to *Catostomus*, *Chasmistes*, *Deltistes*, *Pantosteus*, or *Xyrauchen*).

We identified fossil calibration points through consulting the phylogenetic relationships presented by Bagley et al. (2018) and Chen and Mayden (2012) and the fossil calibration points and dates presented by Bagley et al. (2018); we avoided fossils with uncertain placement and stratigraphy (Table 1). We did not conduct a total-evidence approach to dating because, as previously stated, this approach is misleading with the genera of *Catostomus*, *Chasmistes*, *Deltistes*, and *Xyrauchen*, which are focal to our study. Instead, we dated the most recent common ancestor of sets of taxa by using the fossil calibrations.

To generate the time-calibrated phylogeny, we used MrBayes (Ronquist et al. 2012) with a relaxed clock model and exponentially distributed fossil calibration points. For initial runs, we partitioned mtDNA by codon position, partitioned *irbp* by codon position, and combined introns (*rps7* I, *gh* I). A general time-reversible (GTR) model of nucleotide evolution with gamma distributed rate variation among sites (Γ) with a proportion of invariant sites was specified, and we did not provide a starting tree. Run state was assessed with the sump command of MrBayes and through Tracer (Rambaut et al. 2018). These initial (10 million state) runs showed a lack of convergence and overall low effective sample sizes (ESSs) for tree likelihood values. We then reduced parameterization by creating a three-partition scheme of mtDNA loci, *irbp*, and combined introns. A starting tree was generated with IQ-TREE (Nguyen et al. 2015) under the GTR+Γ model of nucleotide evolution and 1000 bootstraps. The resulting consensus tree was included in the MrBayes infile, and several tree searches were conducted. We required the ESSs for all parameters to be greater than or equal to 200 before stopping. A maximum compatibility tree was generated as the time-calibrated phylogenetic hypothesis that is presented and used for ancestral state reconstructions.

Sampling, DNA sequencing, and alignment

For a thorough evaluation of the genetic relationships and characteristics of the Wall Canyon Sucker, we obtained DNA extracts or tissue samples from the Wall Canyon

TABLE 1 Fossil calibration points, with fossils, sources, most recent common ancestor (MRCA) of species in phylogeny dated by calibration, and the minimum (min.) age and mean age (millions of years) of exponential distributions.

Fossil	Sources	Dating MRCA	Min. age	Mean age
Amyzon aggregatum,[a] *A. brevipinne*[a]	Wilson 1977; Grande et al. 1982; Wolfe et al. 2003; Greenwood et al. 2005	Chinese Sucker *Myxocyprinus asiaticus*, Quillback *Carpiodes cyprinus*, Highfin Carpsucker *Carpiodes velifer*, River Carpsucker *Carpiodes carpio*, Blue Sucker *Cycleptus elongatus*, Southeastern Blue Sucker *Cycleptus meridionalis*, Bigmouth Buffalo *Ictiobus cyprinellus*, Black Buffalo *Ictiobus niger*	48.88	63.54
A. huanensis[a]	Cheng 1962; Chang et al. 2001; Walker et al. 2012	River Carpsucker, Quillback, Highfin Carpsucker, Bigmouth Buffalo, Black Buffalo	37.80	49.14
Catostomus oromyzon[a]	Smith et al. 2013	Yaqui Sucker *Catostomus bernardini*, Cahita Sucker *Catostomus cahita*, Flannelmouth Sucker *Catostomus latipinnis*, Sonora Sucker *Catostomus insignis*, Fleshylip Sucker *Catostomus leopoldi*	4.50	5.85
Chasmistes spatulifer[a]	Miller and Smith 1967; Smith 1975; Ruez and Gensler 2008	Utah Sucker *Catostomus ardens*, Largescale Sucker *Catostomus macrocheilus*, Tyee Sucker *Catostomus tsiltcoosensis*, June Sucker *Chasmistes liorus mictus*	3.00	3.90

[a]Extinct species.

Sucker and other *Catostomus* species of the western Great Basin (the Owens, Warner, and Tahoe suckers) with collections from different water bodies or drainages. As previous work had indicated some similarities between the Wall Canyon Sucker and the Modoc Sucker, we obtained representation of this species as well as the Sacramento Sucker. The other sampled group of suckers that were geographically proximate to the Wall Canyon Sucker was centered in the Klamath Basin and comprised the Klamath Largescale Sucker *Catostomus snyderi*, Klamath Smallscale Sucker *Catostomus rimiculus*, Shortnose Sucker *Chasmistes brevirostris*, and Lost River Sucker *Deltistes luxatus*. We also obtained samples of the Lahontan Sucker *Pantosteus lahontan*, as it is a geographically proximate sucker lineage and could be used for rooting phylogenies of *Catostomus*.

Genomic DNA was extracted from tissue samples with a Qiagen DNeasy extraction kit according to the manufacturer's instructions. The RADseq libraries were generated by following the BestRAD protocol with the *Sbf*1 restriction enzyme (Ali et al. 2016). Libraries were sequenced with 150-base-pair paired-end chemistry on an Illumina HiSeq 4000 at the University of California–Davis Genome Center.

Sequence data files were demultiplexed and then aligned to the Razorback Sucker *X. texanus* (Abbott 1860) reference genome (JAOTQA000000000.1) with the Burrows–Wheeler aligner and the MEM algorithm (Li and Durbin 2009). The sorting of alignments, removal of PCR duplicates, and calculation of the number of aligned reads were performed with SAMtools (Li et al. 2009). Individuals with more than 1,500,000 reads after filtering were downsampled using SAMtools to 1,500,000 reads for analyses.

RADseq phylogeny and network

We selected representative samples of outgroup species (i.e., not the Wall Canyon, Warner, or Owens sucker) from the RADseq data by choosing two individuals from each sample location with the highest number of read counts. We then identified all Wall Canyon, Warner, and Owens Sucker individuals with more than 400,000 filtered and aligned reads and retained those individuals for the phylogenetic analysis. Genotypes were called with ANGSD (Analysis of Next Generation Sequencing Data; Korneliussen et al. 2014) using the following general options to create a PLINK-formatted file: −doPlink 2 and −doGeno 4. We used a SAMtools model (−GL 1) and inferred major and minor alleles from genotype likelihoods (−doMajorMinor 1) with fixed major and minor allele frequencies (−doMaf 1). We also used the following quality control options: the presence of a single-nucleotide polymorphism (SNP) in 95% of individuals (−minInd 113), a minimum minor allele frequency of 0.05 (−minMaf 0.05), minimum mapping quality of 10 (−minMapQ 10), a minimum base quality of 20 (−minQ 20), a SNP *p*-value of 1×10^{-6} (−SNP_pval 1e6), and a posterior cutoff of 0.95 (−postCutoff 0.95).

We converted the resulting PLINK-formatted file to a variant call format (VCF) file with PLINK (Purcell et al. 2007) and removed linked SNPs with BCFtools (+prune −l 0.9 −w 100,000; Danecek et al. 2021). The resulting VCF file was converted to a PHYLIP file (https://github.com/edgardomortiz/vcf2phylip/blob/master/vcf2phylip.py) and prepared from maximum likelihood inference (https://github.com/btmartin721/raxml_ascbias/blob/master/ascbias.py). Individuals with less than 10% missing data were retained for phylogenetic analysis (https://github.com/MacCampbell/relict-dace/blob/master/201.1-removeMissing.pl). A maximum likelihood tree search was conducted with IQ-TREE (Nguyen et al. 2015), implementing ascertainment bias correction and a model search (−m MFP+ASC). The robustness of inferred relationships was assessed with 1000 rapid bootstraps (−bb 1000). Nodes with less than 75% bootstrap support (bs) were collapsed to polytomies.

A concatenated phylogenetic analysis resulting in a bifurcating tree, such as the preceding maximum likelihood analysis, does not accurately represent complex evolutionary scenarios. Evolutionary processes such as incomplete lineage sorting (ILS; deep coalescence) or hybridization can generate a conflict between gene trees and a species tree but can be visualized in a phylogenetic network to generate a more representative depiction of evolutionary relationships (Huson and Bryant 2005; Huson et al. 2010; Solís-Lemus and Ané 2016). We created an implicit phylogenetic network from the same alignment used in the maximum likelihood tree search with a Neighbor-Net algorithm in the program SplitsTree to visualize potential ILS and hybridization across time scales in our phylogenetic alignment (Huson and Bryant 2005).

Population genetic relationships

We analyzed samples of Wall Canyon, Warner, and Owens suckers in a population genetic framework. First, we identified individuals from these three lineages that had at least 90,000 aligned reads after filtering for proper pairing and after removal of PCR duplicates. Next, we conducted principal components analysis and admixture analysis in a genotype likelihood framework. These analyses can provide evidence of recent gene flow and admixture that are not evident or clearly measurable in phylogenetic analyses.

We created a genotype likelihood file with ANGSD (−doGLF 2), with other options following the genotype calls made for phylogenetic analysis except the default

posterior cutoff. We generated a covariance matrix with PCAngsd (Meisner and Albrechtsen 2018) and converted it to coordinates for principal components analysis within R (R Development Core Team 2022). Admixture analyses were conducted with NGSAdmix from two putative genetic clusters ($K = 2$) and upwards until subdivision of individuals within sampling locations indicated overparameterization (Skotte et al. 2013).

Population genetic characteristics

A set of SNPs for statistical analysis from Wall Canyon, Warner, and Owens Sucker individuals with at least 90,000 aligned reads was created by calling genotypes. The same specifications as the phylogenetic genotype calls with ANGSD were used. The PLINK-formatted output was converted to a VCF file with PLINK (Purcell et al. 2007) and then pruned with the +prune algorithm of BCFtools (+prune −l 0.6 −w 10,000; Danecek et al. 2021). We imported the pruned VCF file into R and conducted analyses of it by using the snpR package (https://github.com/hemstrow/snpR). Pairwise values of the genetic differentiation index F_{ST} were calculated using the cacl_pairwise_fst function of snpR between the Wall Canyon, Warner, and Owens suckers across all samples. Subsequently, sampling locations with at least eight individuals were retained and treated individually for additional pairwise F_{ST} calculations. Sampling locations with at least eight individuals were also used in the calculation of nucleotide diversity (π) and Tajima's D. Nucleotide diversity serves as a measure of genetic variation—a metric that is critically important for short- and long-term population viability (Kardos et al. 2021). Tajima's D provides insight into whether a population is evolving neutrally or whether a nonrandom process, such as a population size change, is occurring (Tajima 1989). We calculated π with the calc_pi function and Tajima's D with the calc_tajimas_d function of snpR. A mean estimate for each sampling location for both π and Tajima's D was generated. We tested whether the measurements of π and Tajima's D differed among sampling locations with at least eight individuals by using a Wilcoxon rank-sum test (wilcox.test in R). The Wilcoxon rank-sum test as used in our study is equivalent to the Mann–Whitney test and is a nonparametric test if one distribution is shifted to the left or right of the other.

Molecular species delimitation

We identified *nd2* sequences from GenBank that were representative of the Wall Canyon Sucker, Warner Sucker, Owens Sucker, and Cui-ui. Only one sequence from the Cui-ui was available; therefore, the species was omitted from further molecular species delimitation analysis. Sequences were aligned with MAFFT (Katoh and Toh 2008; Katoh and Standley 2013), and a phylogeny was generated by importing into R and conducting 1000 bootstraps of a neighbor-joining tree with uncorrected distance using functions of the ape (Analyses of Phylogenetics and Evolution) package (Paradis and Schliep 2019). The alignment was then uploaded to the Assemble Species by Automatic Partitioning (ASAP) webserver, specifying uncorrected distances (https://bioinfo.mnhn.fr/abi/public/asap/; Puillandre et al. 2021).

RESULTS

Re-assessment of *Catostomus* relationships and divergence times

An alignment of 51 tips, 5925 characters, and 1970 distinct alignment patterns was analyzed. A sufficient ESS (>596) for all parameters was achieved by combining three runs of four chains per run for 40 million generations sampled every 2000 generations and applying a 10% burn-in as evaluated by Tracer (Rambaut et al. 2018). The maximum compatibility tree was generated with the sumt command of MrBayes (sumt relburnin = yes burninfrac = 0.10 nruns = 3 contype = allcompat outputname = allcompat). The MrBayes input nexus file, summarized parameter files, and final maximum compatibility tree are provided at https://doi.org/10.5281/zenodo.7640819. The time-calibrated tree is presented as Figure S1 (available in the Supplement in the online version of this article), with a subtree of the Wall Canyon Sucker and near-relatives presented as Figure 2. In the time-calibrated tree figures, the HPD of a node age is presented as a horizontal bar, and the estimated node age is presented as a posterior median.

A clade consisting of the Wall Canyon Sucker, Warner Sucker, Cui-ui, and Owens Sucker received maximal support, with a posterior probability of 1.00 for all nodes in that clade (henceforth referred to as "Wall Canyon Sucker and relatives"). Within this clade, the Wall Canyon Sucker was sister to the Warner Sucker, with an estimated divergence time of 2.12 million years (95% HPD = 0.45–4.13). The time to most recent common ancestor (TMRCA) of this clade was 9.40 million years (95% HPD = 5.86–13.28). The Cui-ui and Owens Sucker were indicated to be most closely related, with a TMRCA of 4.96 million years (95% HPD = 2.17–8.19).

The Wall Canyon Sucker and relatives were found to be sister to a lineage of suckers split broadly between the Sacramento River basin and the Klamath River basin

(Figure 2). The support for branching relationships and monophyly of the three main clades was maximal.

Sampling, DNA sequencing, and alignment

We obtained 307 sucker samples for sequencing that averaged 1,306,545 reads/sample after filtering and removal of PCR duplicates (Tables 2 and S1 [available in the Supplement in the online version of this article]). We sequenced 27 Wall Canyon Sucker individuals from several sites along Wall Creek. From the potential sister species, we sampled 38 Warner Suckers from three locations and 44 Owens Suckers from four locations (Figure 1B).

RADseq phylogeny and network

We selected 119 samples for genotype calling based on two samples with the highest read counts per sampling location

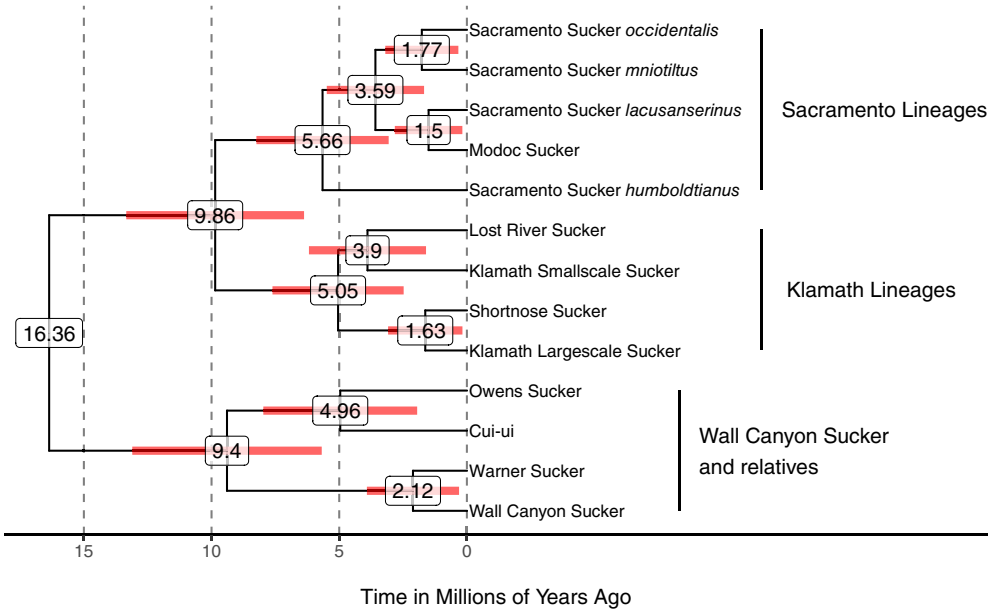

FIGURE 2 Subtree of time-calibrated phylogeny generated using a four-gene alignment. Posterior median age estimates (millions of years ago) are presented at nodes, with 95% highest posterior densities shown as horizontal bars. The Wall Canyon Sucker and the Warner Sucker are most closely related to each other, with an estimated divergence of 2.12 million years ago. The Owens Sucker and Cui-ui as a combined lineage are the sister lineage of the Warner and Wall Canyon suckers. Sacramento Sucker *Catostomus occidentalis* subspecies names are appended to tips.

TABLE 2 Species common name (as identified by collectors), scientific name, sample size (*N*), and average number of filtered reads for each species.

Species common name	Species scientific name	*N*	Average number of aligned and filtered reads
Wall Canyon Sucker	*Catostomus* sp.	27	1,622,606
Warner Sucker	*Catostomus warnerensis*	38	612,234
Owens Sucker	*Catostomus fumeiventris*	44	2,502,955
Modoc Sucker	*Catostomus microps*	20	353,922
Sacramento Sucker	*Catostomus occidentalis*	27	259,208
Klamath Smallscale Sucker	*Catostomus rimiculus*	16	1,335,915
Klamath Largescale Sucker	*Catostomus snyderi*	25	797,978
Unidentified sucker	*Catostomus* sp.	3	625,320
Tahoe Sucker	*Catostomus tahoensis*	28	1,686,146
Shortnose Sucker	*Chasmistes brevirostris*	25	851,760
Lost River Sucker	*Deltistes luxatus*	25	851,760
Lahontan Sucker	*Pantosteus lahontan*	29	1,844,688

if not Wall Canyon, Warner, or Owens Suckers (n = 26). We then included all samples with over 400,000 aligned reads after filtering from Wall Canyon Suckers (n = 27), Warner Suckers (n = 22), and Owens Suckers (n = 44). These 119 samples produced an initial 37,855 SNPs. Twelve samples were removed due to a high proportion of missing data, producing a final alignment of 107 samples and 8168 SNPs with 8066 distinct alignment patterns and 7330 parsimony-informative sites. The 12 samples removed were two Lahontan Suckers (Poore Creek sampling location), two Sacramento Suckers (Ash Creek sampling location), two Modoc Suckers (Ash Creek sampling location), and six Warner Suckers (n = 2 from the Deep Creek sampling location; n = 4 from the Twentymile Creek sampling location).

The phylogenetic alignment contained four Lahontan Suckers: two from the Carson River and two from the Truckee River. It also contained four Tahoe Suckers: two from the Carson River and two from the Truckee River. Two Goose Lake Suckers *Catostomus occidentalis lacusanserinus* (a Sacramento Sucker subspecies) from the Cottonwood sampling location and two Modoc Suckers from the Turner Creek sampling location remained after selection of high-read-count individuals and filtering for missing data. As specific locality information was unavailable for Klamath Largescale, Klamath Smallscale, Shortnose, and Lost River suckers, two individuals of each species were present in the phylogenetic alignment.

The consensus tree is presented as Figure 3, and the alignment and output files from IQ-TREE are provided at https://doi.org/10.5281/zenodo.7640819. Backbone nodal supports were maximal, with bs of 100%, as well as support for the monophyly of placement as sister lineages of the Wall Canyon and Warner suckers. No apparent structuring was observed within the Wall Canyon Sucker. The Warner Sucker samples separated into sampling locations: Honey Creek (bs = 100%) was sister to a single Deep Creek sample (bs = 95%) and a separate clade of Twentymile Creek individuals (bs = 100%). The Owens Sucker was divided into two well-supported clades, with the Lower Hot Creek sampling location being monophyletic (bs = 100%) and all other sampling locations being combined into a clade (bs = 92%).

The implicit network, presented as Figure S2, demonstrated a major reticulation between the Sacramento Sucker and the Modoc Sucker. There was some evidence of ILS or hybridization within sucker species from the Klamath Basin. The Wall Canyon, Warner, and Owens suckers were shown to be most closely related.

Population genetic relationships

The combined population genetics data set of Wall Canyon (n = 27), Warner (n = 37), and Owens (n = 44) suckers was composed of 108 fish in total. The principal components and admixture analyses were conducted on a set of 153,443 genotype likelihoods. The first principal component (PC1) separated the Owens Sucker from the other two suckers (56.82% of the variance), and PC2 distinguished the Wall Canyon Sucker from the Warner Sucker (14.64% of the variance; Figure 4A). The third PC separated Owens Suckers between the northern (Lower Hot Creek) and southern (other localities) sampling locations in this study (1.64% of the variance; Figure 4B). A scree plot supporting the presentation of three PCs is provided in Figure 4C. Admixture analyses were performed for K-values from 2 to 6 and showed evidence of overparameterization at a K of 6; therefore, results for K-values from 2 to 5 are presented (Figure 5). At a K-value of 2, the first division separated the Wall Canyon and Warner suckers from the Owens Sucker. With additional K-values, the Wall Canyon, Warner, and Owens suckers formed distinct clusters (K = 3). At a K-value of 4, the Owens Sucker was subdivided into a Lower Hot Creek genetic cluster and all other sampling locations. At a K of 5, the Warner Sucker was subdivided between the Honey Creek and Twentymile Creek sampling locations, with Deep Creek individuals being mixed in ancestry.

Population genetic characteristics

Pruning of a total of 22,200 called SNPs with the +prune algorithm in BCFtools resulted in 9511 SNPs for analysis (VCF file is provided at https://doi.org/10.5281/zenodo.7640819). Pairwise F_{ST} values between putative species were 0.14 between the Wall Canyon and Warner suckers, 0.18 between the Wall Canyon and Owens suckers, and 0.18 between the Warner and Owens suckers. Pairwise F_{ST} measurements between sampling locations are presented in Table 3. Values of F_{ST} comparing the Wall Canyon Sucker to the Warner Sucker across sampling locations ranged from 0.15 to 0.21, and F_{ST} measurements comparing the Wall Canyon Sucker to the Owens Sucker across sampling locations were 0.20 and 0.22. Values of F_{ST} comparing the Warner Sucker to the Owens Sucker across sampling locations ranged from 0.16 to 0.22. Intraspecific measures of F_{ST} were between 0.02 and 0.06 for the Warner Sucker and 0.08 for the Owens Sucker.

Estimates of π were lowest in the Wall Canyon Sucker (0.13), intermediate in the Warner Sucker (0.15–0.16), and highest in the Owens Sucker (0.18–0.20; Figure 6A). The center of the Wall Canyon Sucker distribution of π was less than the center of all other distributions of π (Table S2; p-values = 8.41×10^{-24} to 3.74×10^{-154}). Estimates of Tajima's D were all greater than 0; values were highest in the Wall Canyon Sucker (1.96) and were lower in both the Warner

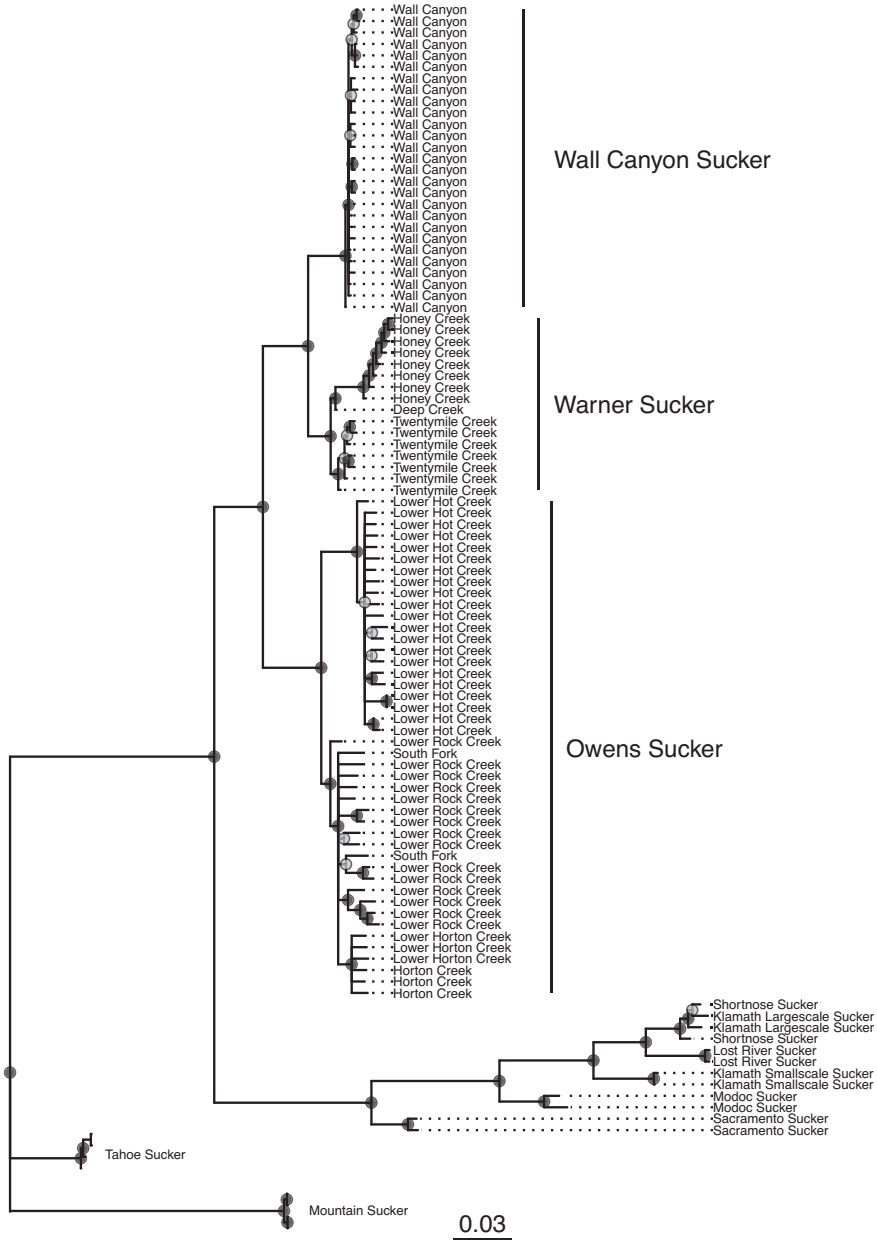

FIGURE 3 Maximum likelihood phylogeny generated from an alignment of 8168 single-nucleotide polymorphisms. The bootstrap consensus tree is presented, with bootstrap support values less than 75% collapsed into polytomies. Bootstrap values of at least 75% but less than 90% are indicated with light gray circles; values of 90% or greater are indicated with dark gray circles. For outgroup species, the common name from field identification is given (see Table 2 for scientific names). For the Wall Canyon, Warner, and Owens suckers, clades are labeled, with collection locations of individuals supplied at tips.

Sucker (0.63–1.30) and the Owens Sucker (1.31–1.33; Figure 6B). In all comparisons, the Wall Canyon Sucker had a distribution of Tajima's *D* centered at a higher value than all other distributions of Tajima's *D* (Table S2; *p*-values = 1.09×10^{-11} to 9.92×10^{-12}).

Molecular species delimitation

We created an alignment of nine *nd2* sequences—three each from the Wall Canyon Sucker, Warner Sucker, and Owens Sucker. The resulting alignment was 1047 base pairs in length and had 92 segregating sites and no gaps. The neighbor-joining tree of *nd2* sequences identified three well-supported clades (bs ≥ 98%) corresponding to the Wall Canyon, Warner, and Owens suckers (Figure S3). The ASAP algorithm predicts three species in the scenario with the highest support, corresponding to the Wall Canyon, Warner, and Owens suckers. Output from the ASAP webserver, including scores, ranking threshold distances, and a SPART (standardized species partition) formatted file, is available at

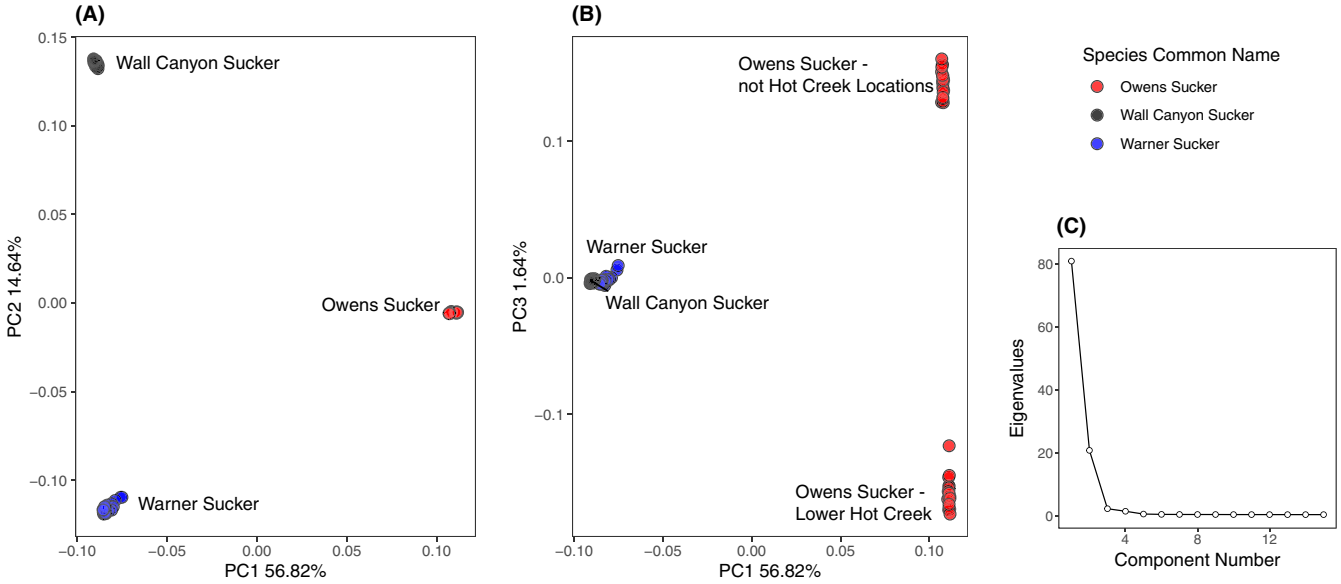

FIGURE 4 (A), (B) Principal components analysis of the Wall Canyon, Warner, and Owens suckers for the first three principal components (PC1–PC3), with the percentage of variance explained by each PC indicated; and (C) the associated scree plot.

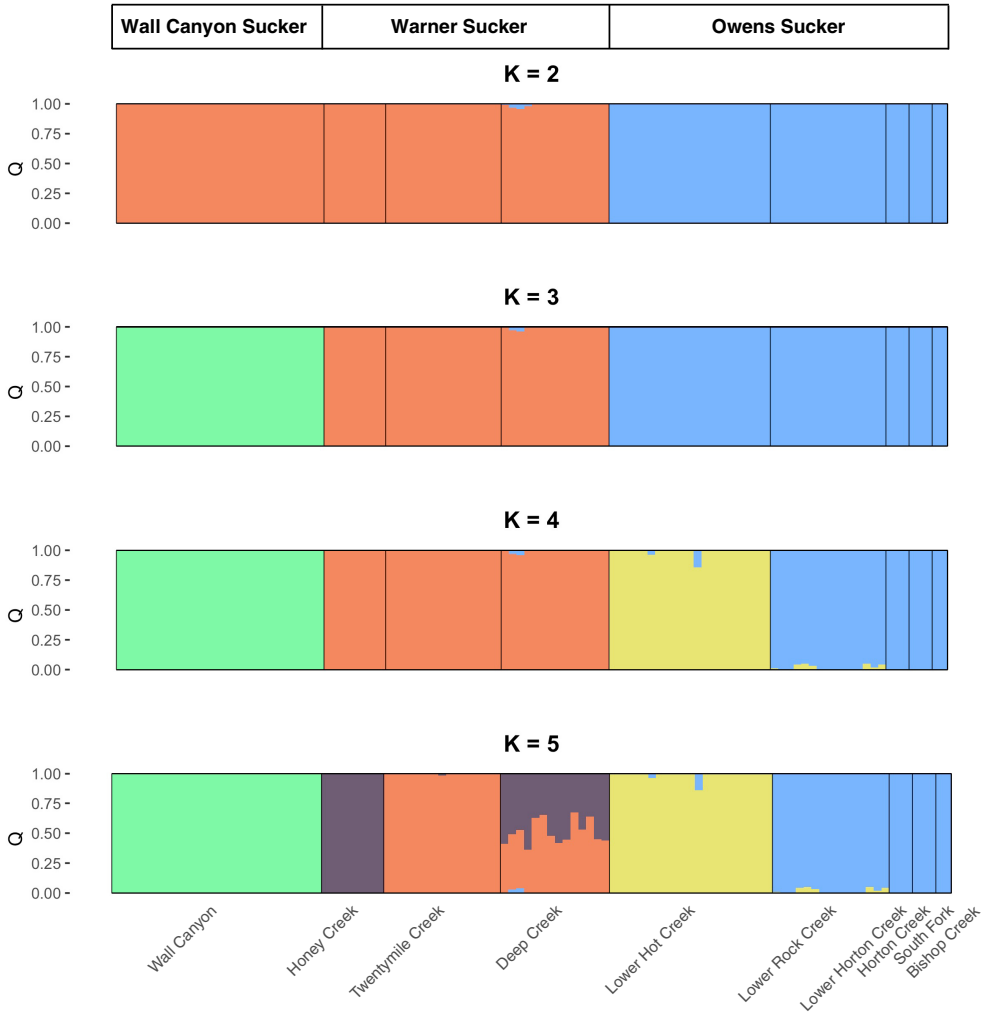

FIGURE 5 Admixture analyses using *K*-values of 2–5 (i.e., number of putative genetic clusters) for the Wall Canyon, Warner, and Owens suckers. Sampling locations are indicated on the *x*-axis, and the ancestry coefficient (*Q*) is indicated on the *y*-axis.

TABLE 3 Pairwise genetic differentiation index F_{ST} measurements of Wall Canyon, Warner, and Owens suckers from sampling locations with at least eight individuals.

		Warner Sucker			Owens Sucker	
Species	**Location**	**Honey Creek**	**Twentymile Creek**	**Deep Creek**	**Lower Hot Creek**	**Lower Rock Creek**
Wall Canyon Sucker	Wall Canyon	0.21	0.17	0.15	0.20	0.22
Warner Sucker	Honey Creek	–	0.06	0.04	0.22	0.20
	Twentymile Creek		–	0.02	0.19	0.18
	Deep Creek			–	0.17	0.16
Owens Sucker	Lower Hot Creek				–	0.08

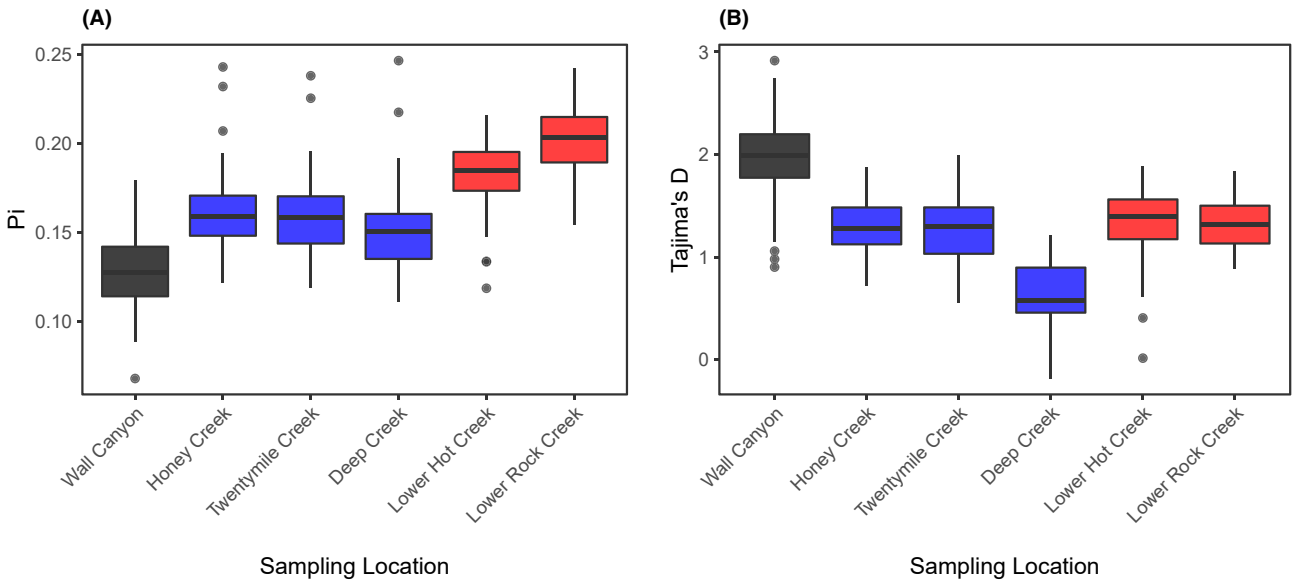

FIGURE 6 (A) Nucleotide diversity (π) and (B) Tajima's *D* presented for sampling locations with at least eight individuals of the Wall Canyon Sucker (Wall Canyon), Warner Sucker (Honey, Twentymile, and Deep creeks), and Owens Sucker (Lower Hot and Lower Rock creeks). Nucleotide diversity and Tajima's *D* are presented as boxplots with medians indicated by a horizontal line, the interquartile range from the first to third quartiles as boxes with whiskers extending 1.5 times the interquartile range. Outliers beyond the whiskers are shown as points.

https://doi.org/10.5281/zenodo.7640819 along with the multiple sequence alignment.

DISCUSSION

Our molecular phylogenetic re-assessment of *Catostomus* relationships and divergence times placed the Wall Canyon Sucker in a clade containing two genera (*Catostomus* and *Chasmistes*). We did find that the Warner Sucker appears to be the nearest extant relative of the Wall Canyon Sucker, as was previously shown. Our divergence time was substantially less than the only comparable estimate from Bagley et al. (2018), most likely as a result of not using a total-evidence approach. Our approach estimated divergence times on a topology generated from molecular sequence data demonstrating the non-monophyly of *Catostomus* and included a key lineage that was not previously indicated to be a close relative of the Wall Canyon Sucker—the Cui-ui (a member of the genus *Chasmistes*). The lower 95% HPD of Bagley et al. (2018) for the divergence of the Wall Canyon Sucker and the Warner Sucker was pre-Pleistocene (~2.58 mya), whereas we found a lower 95% HPD of approximately 0.45 mya, with a posterior median of 2.12 mya. Thus, at a minimum, our estimate of divergence is 2 million years younger than that of Bagley et al. (2018). Our finding does indicate meaningful divergence between the Wall Canyon Sucker and the Warner Sucker (2.12 mya).

Substantial representation of the Wall Canyon Sucker through RADseq resulted in reciprocal monophyly of both the Wall Canyon Sucker and the Warner Sucker with high support (Figure 3). A phylogenetic network does not show evidence of gene tree and species tree

discordance arising from ancient or recent hybridization, for example, that would mislead the placement of the Wall Canyon Sucker (Figure S2). We detected no signal of recent admixture between the Wall Canyon and Warner suckers (Figures 4 and 5). Divergence of the Wall Canyon Sucker from the Warner Sucker in terms of F_{ST} exceeded the observed intraspecific divergences in this study. In the case of the principal components analyses, the variance explained by PC2 (14.64%), which separated the Wall Canyon and Warner suckers, was much greater than the variance explained by PC3 (1.64%), which separated genetic clusters of Owens Suckers (Figure 4). Additionally, PC4, with 1.07% of the variance, separated Warner Sucker sampling locations (results not shown). Correspondingly, there were substantially high F_{ST} values (~0.16) between the Wall Canyon Sucker and other species when compared with intraspecific measures of F_{ST} (0.02–0.08) in this study. These lines of evidence point to clear isolation between the Wall Canyon Sucker and the Warner Sucker and the independence of those lineages, which can be interpreted as a species-level divergence. Further examination of the Wall Canyon Sucker with molecular species delimitation using *nd2* sequences identified the Wall Canyon Sucker as a species.

Population genetic characteristics of the Wall Canyon Sucker compared to closely related species indicated that it had the lowest π and the greatest Tajima's D (~2.00). In general, Tajima's D was positive across the Wall Canyon, Warner, and Owens suckers (0.63–1.96). Interpretation of these values indicated that there is an overall lack of rare alleles consistent with population contraction. Indeed, the Wall Canyon Sucker's reduction in range can be characterized as extreme, from Lake Surprise with 1366 km^2 of area with additional tributaries approximately 15,000 years ago (Ibarra et al. 2014) to about 38 km of suitable post-pluvial stream habitat within the Wall Canyon drainage and further to 8.7 km of stream habitat occupied at present. Such a large value of Tajima's D could be related to both the contraction of post-pluvial habitat from natural processes as well as more recent exclusion from available habitat by introduced fish (e.g., Brown Trout *Salmo trutta*).

Knowledge of biodiversity and subsequent conservation actions are frequently hampered by major shortfalls in knowledge, such as Wallacean (i.e., distribution; Lomolino 2004), Prestonian (i.e., abundance and spatial distribution; Cardoso et al. 2011), Darwinian (i.e., evolutionary processes and phylogenetic information; Diniz-Filho et al. 2013), and Linnean (i.e., a lack of taxonomic recognition; Brown and Lomolino 1998). For the Wall Canyon Sucker, the Wallacean and Prestonian shortfalls have been overcome by survey data (i.e., 8.7 km of habitat; ~453 individuals), and this study provides insights into the phylogeny and relationships of the Wall Canyon Sucker and related species as well as data for further investigations, overcoming much of the Darwinian shortfall. Lack of taxonomic recognition (the Linnean shortfall) remains an issue for this species.

CONCLUSIONS

A species description for the Wall Canyon Sucker requires the proposal of a taxonomic name; a designation of representative specimens (types) accessioned in a natural history collection; a diagnosis that separates it from other taxa at the same rank or, at a minimum, the Warner Sucker; and a description of representative specimens (e.g., Dubois 2010). The genetic data presented here can form part of the diagnosis, distinguishing the Wall Canyon Sucker from other species of *Catostomus*. Other components of the diagnosis may be a distinct arrangement of the suborbital lateralis canal and its smaller scales in comparison with the Warner Sucker (Markle and Harris 2002). Lateral line scale counts for the Wall Canyon Sucker are 88–102 ($n = 8$), compared to 64–82 ($n = 13$) for the Warner Sucker.

Often, existing taxonomic names are available for use with previously consolidated species when differentiation is highlighted by molecular data, as in this study (e.g., Moyle and Campbell 2022; Wright et al. 2022). There is, however, no published species description, and there is no available name to apply to this species. The Wall Canyon Sucker continues to await formal recognition as a species after nearly 100 years.

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ETHICS STATEMENT

Tissue samples or DNA for this project was obtained from individuals listed in the acknowledgements. Collections for this study followed institutional guidelines and permitting.

DATA AVAILABILITY STATEMENT

Sequence data generated for this study are available from the National Center for Biotechnology Information's Sequence Read Archive under BioProject PRJNA868046. Code for analysis, generation of figures, sequence alignments, and intermediate files are available at https://github.com/MacCampbell/TAFS-wall-canyon and https://doi.org/10.5281/zenodo.7640819.

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SUPPORTING INFORMATION

Additional supplemental information can be found online in the Supporting Information section at the end of this article.